

Bridging Modality Gap for Effective Multimodal Sentiment Analysis in Fashion-related Social Media

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Abstract

Multimodal sentiment analysis for fashion-related social media is essential for understanding how consumers appraise fashion products across platforms like Instagram and Twitter, where both textual and visual elements contribute to sentiment expression. However, a notable challenge in this task is the modality gap, where the different information density between text and images hinders effective sentiment analysis. In this paper, we propose a novel multimodal framework that addresses this challenge by introducing pseudo data generated by a two-stage framework. We further utilize a multimodal fusion approach that efficiently integrates the information from various modalities for sentiment classification of fashion posts. Experiments conducted on a comprehensive dataset demonstrate that our framework significantly outperforms existing unimodal and multimodal baselines, highlighting its effectiveness in bridging the modality gap for more accurate sentiment classification in fashion-related social media posts.

1 Introduction

Fashion products are characterized by significant variability, primarily due to the rapidly changing preferences of consumers (Bilinska, 2021). Users often express their sentiments towards these fashion products through various social media platforms like Instagram and Twitter. Thus, it is important to properly capture users' sentiments about fashion products on social media.

Traditional sentiment analysis tasks for social media mainly focus on text comprehension (Hutto and Gilbert, 2014; Balahur, 2013). However, as visual content has gradually become an important medium for emotional expression, it is crucial to jointly consider information from different modalities to accurately identify the user's real attitude

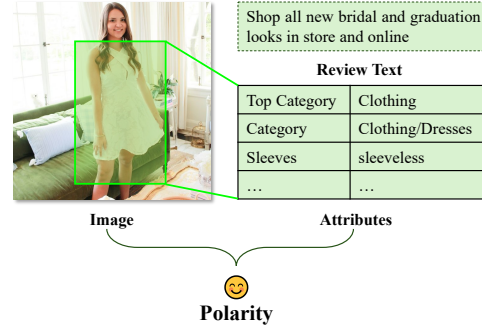


Figure 1: An example of fashion post.

towards fashion items in social media posts (Al-Tameemi et al., 2024). For instance, as illustrated in Figure 1, one must examine the image as well as text content to determine the user's sentiment polarity towards the fashion items. Therefore, our approach focuses on analyzing both textual and visual information to effectively identify sentiment in fashion-related posts.

However, it is a non-trivial task due to the following challenges: The primary challenge is the modality gap, as highlighted by Liang et al. (2022) and Hazarika et al. (2020). The modality gap between text and images stems from the difference in information density: text conveys sentiment directly through high-density emotional language, while images, often filled with much non-sentiment information like environmental details, carry a lower density of sentiment-related information. This difference can affect the model's ability to effectively combine both modalities. (Wei et al., 2023).

Additionally, fashion-related posts often include unique attributes such as style, pattern and color, as shown in Figure 1, that significantly influence sentiment. How to effectively utilize their relationships with text and images is essential.

Early studies, such as Xu and Nan (2017) and Xu and Mao (2017) mainly focus on how to extract

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features from both visual and textual modalities while ignoring the modality gap. In this study, we propose a novel multimodal framework to capture information from both textual and visual modalities for sentiment classification of fashion posts. Specifically, we propose a two-stage framework to generate pseudo data including pseudo images and their corresponding attributes according to the text, representing the textual information in a visual format. Compared with original ones, pseudo images have higher information density, focusing on primary content with minimal background noise. This medium information density of pseudo data helps to narrow the modality gap during the fusion of text and images and improves the accuracy of sentiment prediction.

Subsequently, we design a multimodal fusion approach that efficiently integrates features from different modalities. In our approach, attributes serve to guide the model in capturing essential image information, further narrowing the modality gap. Additionally, we develop a self-adaptive adapter to evaluate the consistency between original and pseudo data, preventing inaccurately generated pseudo data from influencing sentiment prediction.

Experimental results demonstrate that our proposed model outperforms several strong unimodal and multimodal baselines. Additional experiments further prove that the modalities and the designed fusion architecture contribute to the ideal classification results. We also compare our pseudo data generation methods with others, showing that our method can better bridge the modality gap. Our approach not only enhances the accuracy of sentiment analysis but also paves the way for a more comprehensive understanding of user sentiment in the fashion domain.

The main contributions of this work can be summarized as follows:

- We propose a LLM-based framework that generates pseudo data to enrich the original text and bridge the modality gap.
- A novel fusion framework is proposed, incorporating a self-adaptive adapter to filter inaccurate pseudo data.
- Experiments on a fashion-related dataset demonstrate that our model significantly outperforms state-of-the-art unimodal and multimodal baselines.

2 Related Work

In this section, we introduce two related topics about this study: sentiment analysis and fashion-related tasks.

2.1 Sentiment Analysis

Traditional sentiment analysis primarily relies on text to extract sentiment features using natural language processing and machine learning models (Liu, 2020; Poria et al., 2020; Yadav and Vishwakarma, 2020). Notable advancements include multi-task models (Liao et al., 2021) and data augmentation (Li et al., 2023a). Recent applications of large language models also show promising results (Kheiri and Karimi, 2023).

The emerging field of multimodal sentiment analysis combines text, images, and audio to enrich sentiment detection (Zhu et al., 2022; Yu et al., 2023). Approaches like AoBERT (Kim and Park, 2023) unify text, visual, and audio modalities within a single framework, while TEDT (Wang et al., 2023) employs a transformer-based network to synchronize and fuse data across modalities, enabling effective sentiment prediction.

2.2 Fashion Related Tasks

Fashion-related research primarily focuses on fashion image analysis (Ge et al., 2019; Liu et al., 2016; Tian et al., 2023), recommendation systems (Ding et al., 2023; De Divitiis et al., 2023; Dahunsi et al., 2024), and clothing retrieval (Miao et al., 2020; Zhang et al., 2020; Ning et al., 2022), which are crucial for both academy and industry.

The emergence of multimodal techniques in fashion has induced novel applications. Baldrati et al. (2024) enhance the generative editing of fashion images using multimodal prompts, while Singh and Patras (2024) apply vision-language models to support creative fashion design processes. Additionally, Wu et al. (2022) advances recommendation systems by utilizing multimodal data for enriched user interaction.

The research by Yuan and Lam (2022) is particularly relevant to our study as it also focuses on sentiment detection in fashion posts, utilizing a multimodal approach that integrates features from text, images, and attributes for classification. We build upon their foundational dataset and extend their work by further enhancing the integration methods and addressing the modality gap more effectively in our multimodal framework.

3 Method

In this study, our goal is to detect the *sentiment polarity* of fashion-related social media posts. Each post, which includes an image, text, and attributes, is classified into one of three distinct labels: positive, neutral, and negative.

As shown in Figure 2, we propose a multimodal framework that captures and integrates information from both textual and visual modalities for sentiment classification. First, we propose a two-stage framework to generate pseudo data, including pseudo images and their associated attributes. This pseudo data serves as a bridge between the text and image modalities to mitigate the gap between two modalities and enhance the performance of the multimodal model. Next, we propose a multimodal fusion network where attributes serve to guide the representation of images and a self-adaptive adapter is incorporated to filter the erroneously generated pseudo data. The final fusion seamlessly integrates the information across modalities for sentiment classification of fashion posts.

3.1 Pseudo Data Generation

To bridge the gap between text and image modalities, we propose a two-stage framework for generating pseudo data, including pseudo images and their corresponding attributes. As illustrated in Figure 3, the first stage utilizes a large language model¹ to generate pseudo image descriptions and pseudo attribute descriptions. The model produces image descriptions that closely align with the given textual input, while the pseudo attributes provide precise details about the fashion items mentioned pseudo image description.

During the generation process, a prompt strategy is applied to guide the model toward aligning the desirable format. Specifically, it directs the LLM to generate appropriate pseudo image descriptions (*Command 1-2*), produce pseudo attributes that accurately describe fashion elements (*Command 3*), and standardize the output format for subsequent processing (*Command 4-5*).

In the second stage, these pseudo image descriptions are used to generate the corresponding pseudo images using Stable Diffusion (Rombach et al., 2022).

This process results in the generation of pseudo images and detailed pseudo attribute descriptions, which can be used in subsequent steps.

¹GPT-3.5 Turbo

3.2 Text and Attributes Representation

The encoding process for text, attributes, and pseudo attributes representations begins by tokenizing the input m , where $m \in \{T, A, P_A\}$. Here, T, A, P_A separately denote text, attributes, and pseudo attributes. These tokens are first transformed into token embeddings. These embeddings are then augmented with positional embeddings to incorporate sequence information:

$$z_m[i] = \text{TextTokenizer}(\text{token}[i]) + E_{\text{pos}}[i] \quad (1)$$

where E_{pos} represents the positional encodings that integrate sequence information into the embeddings and $m \in \{T, A, P_A\}$.

The initial embeddings z_m are then processed by a CLIP (Radford et al., 2021) text encoder (Transformer architecture), which utilizes self-attention mechanisms and feed-forward networks to generate a comprehensive text representation:

$$H_m = \text{Encoder}(z_m)(m \in \{T, A, P_A\}) \quad (2)$$

Here, H_m signifies the high-level text representation synthesized by the encoder.

3.3 Image Representation

We then employ CLIP (Radford et al., 2021) image encoder to learn the original and pseudo image representation. This encoder consists of a ViT (Dosovitskiy et al., 2020) model, which begins by processing the input image I and pseudo image P_I through resizing and normalizing. The image is then segmented into patches of size $P \times P$, and each patch is flattened and transformed through a linear projection:

$$\text{patch}_{\text{embed}}[i] = \text{Flatten}(\text{patch}[i]) \cdot E \quad (3)$$

where E denotes the projection matrix of dimensions $(P^2 \cdot C) \times D$, transforming each patch into a D -dimensional embedding. (C refers to the number of channels)

To incorporate spatial information, positional encodings E_{pos} are added to the embeddings:

$$z_m[i] = \text{patch}_{\text{embed}}[i] + E_{\text{pos}}[i] \quad (4)$$

These embeddings z_m $m \in \{I, P_I\}$ are subsequently processed by the Transformer encoder, utilizing layers of multi-headed self-attention and feed-forward networks to synthesize a comprehensive image representation:

$$H_m = \text{ViTEncoder}(z_m)(m \in \{I, P_I\}) \quad (5)$$

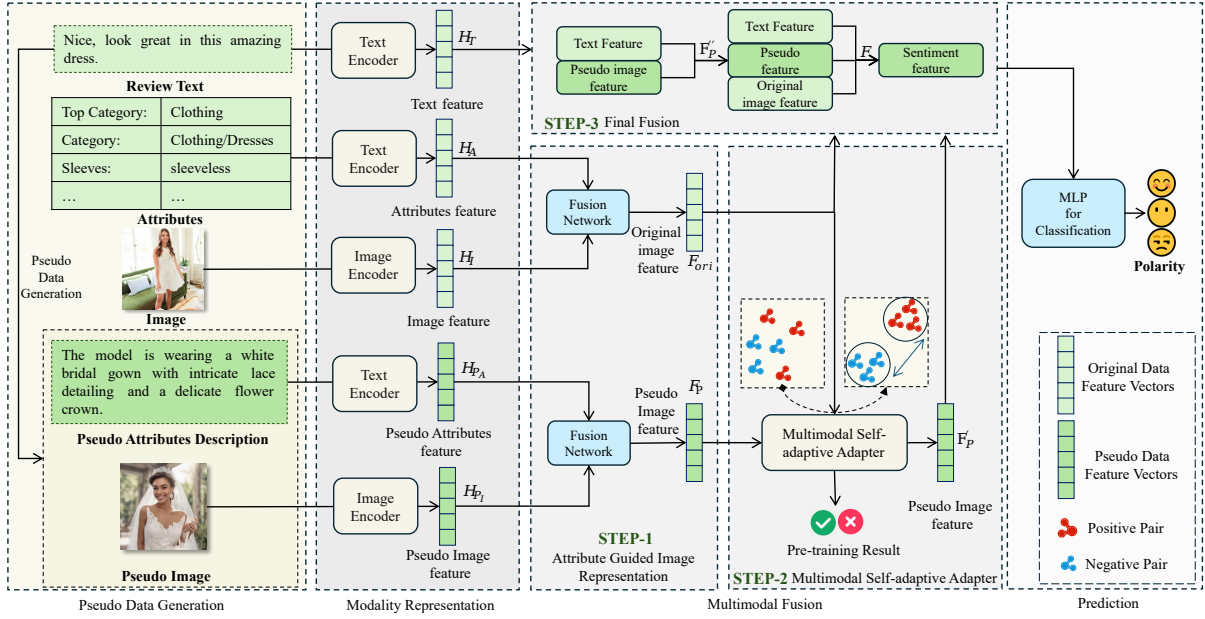


Figure 2: The overview of proposed model.

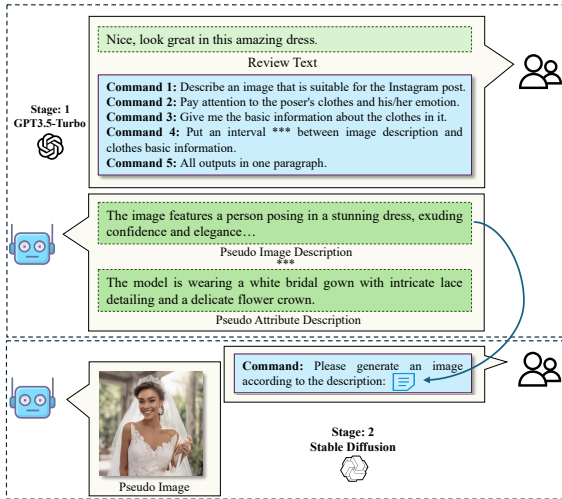


Figure 3: Pseudo data generation.

3.4 Multimodal Fusion

Our multimodal fusion strategy aims to effectively handle the relationships between modalities and further narrow the modality gap. This part includes three main steps: Attribute Guided Image Representation, Multimodal Self-adaptive Adapter and Final Fusion. In our work, the basic fusion network includes two MLP layers and a concatenation.

Attribute Guided Image Representation

The attributes refer to fashion items in images that are the focus of customers' sentiments. Therefore, we use attribute information to guide the representation of images by fusing them to let our model

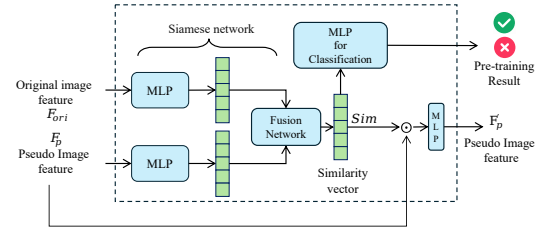


Figure 4: Self-adaptive adapter.

capture the essential information in images, narrowing the modality gap by reducing the redundant information in images. The fused vector, denoted as F_{ori} , is computed as follows:

$$F_{ori} = \text{Fusion}(H_I, H_A) \quad (6)$$

Here, H_I and H_A represent the encoded image and attribute features, respectively. The vector F_{ori} serves a dual purpose: it is fed into a self-adaptive adapter for further processing, and also utilized as the original image feature for sentiment prediction.

The pseudo information fusion is similar to the fusion of the original images and their attributions. We tokenize and encode the generated images and attributes with the CLIP model and then fuse them into pseudo image feature F_P .

Multimodal Self-adaptive Adapter

We then utilize a multimodal self-adaptive adapter to filter out inaccurately generated pseudo images and attributes. Specifically, we aim for the pseudo information to be highly integrated into the final

fusion when it exhibits strong consistency with the original information. Conversely, when the pseudo information conflicts with the original one, we seek to minimize its contribution. To achieve this, we design a pre-training task that assesses the consistency between two input features and projects the consistent and inconsistent fusion information into distinct vector spaces.

The detailed architecture of this component is illustrated in Figure 4. Within this framework, a Siamese network composed of two MLP layers and a fusion network processes the pseudo image feature alongside the original image feature F_{ori} derived from the main backbone, subsequently outputting their similarity feature:

$$Sim = SN(F_{ori}, F_P) \quad (7)$$

where SN represents a Siamese network.

The similarity feature indicates whether the pseudo data matches the original data and decides the intensity of intervention of pseudo information in the final fusion. To obtain such abilities of the similarity, we use dot product to fuse the similarity vector (Sim) with the generated image feature (F_P) as:

$$F'_P = Sim \circ F_P \quad (8)$$

Where F'_P represents the adapted pseudo image feature, the ultimate pseudo information which is retrieved by the main backbone.

Final Fusion

For sentiment prediction, we first fuse F'_P captured from the multimodal self-adaptive adapter with textual feature by:

$$F''_P = \text{Fusion}(F'_P, H_T) \quad (9)$$

Where F''_P represents the sentiment feature extracted from pseudo inputs.

Then we obtain the integral feature by fusing the original information, containing the image and text feature, with pseudo sentiment feature for reference by:

$$F = \text{Fusion}(F''_P, H_T, F_{ori}) \quad (10)$$

This strategy aids in the independent processing and optimization of diverse data types, thereby enhancing the model's flexibility and accuracy, reducing error propagation, and significantly improving the model's generalizability.

This multimodal fusion approach not only facilitates a comprehensive sentiment analysis by

	Positive	Negative	Neutral	Total
Train	3487	2215	3947	9649
Val	462	263	481	1206
Test	438	249	519	1206

Table 1: Statistic of the dataset.

leveraging diverse data types but also addresses the challenges posed by the inherent difference in modality-specific features, ensuring robustness and accuracy in our sentiment classification framework.

3.5 Training Procedure

We first design a pre-training task to provide the self-adaptive adapter module with prior knowledge about the consistency of different image representations. Since sentiment polarity is the primary focus of our task, we emphasize sentiment consistency during the pre-training phase. Specifically, samples in the training set are shuffled and paired, with pairs labeled as "true" if their sentiments match and "false" otherwise. In this step, the self-adaptive adapter is trained to classify the constructed pre-training dataset based on these labels. In this step, all modules except the final fusion are trained.

Next, the entire model is fine-tuned for sentiment analysis. This phase uses insights from the self-adaptive adapter to improve the model's sentiment classification capabilities. The fine-tuning utilizes the cross-entropy loss function:

$$H(p, q) = - \sum_{i=1}^C p_i \log q_i \quad (11)$$

where C is the number of classes, p_i represents the actual distribution, and q_t is the predicted probability for each class. In this step, we finetune the whole model.

4 Experiments

In this section, we present some experimental details, including datasets, evaluation metrics, baseline models, and experimental results.

4.1 Data and Setting

The dataset we use, released by [Yuan and Lam \(2022\)](#), contains over 12,000 fashion-related social media posts. Each post includes an image, fashion attributes, and accompanying text. In the dataset, sentiment labels were manually annotated, considering both the image and text content. The

Modality	Method	Model	ACC.	P.	R.	F1.
Uni-Modality	Language Model	T5 (Raffel et al., 2020)	72.02	72.91	73.12	72.51
		LLaMA-3 (Dubey et al., 2024)	74.37	74.61	74.37	74.27
		BERT (Devlin et al., 2018)	72.88	72.75	75.02	71.32
		Xlnet (Yang et al., 2019)	71.83	65.54	64.47	63.02
		DeBERTa (He et al., 2020)	74.81	74.78	74.88	71.50
	Vision Model	RepVGG (Ding et al., 2021)	42.49	34.38	33.42	21.62
		ResNet18 (He et al., 2016)	48.21	48.94	52.27	47.83
		VIT (Dosovitskiy et al., 2020)	55.10	54.81	54.55	51.96
Multi-Modalities	Related Work	CLMLF (Li et al., 2022b)	74.62	75.21	74.87	74.02
		BIT (Xiao et al., 2023)	67.56	68.09	67.13	67.58
		Yuan’s (Yuan and Lam, 2022)	71.59	70.91	72.17	70.88
	Ours			76.64	76.79	77.62

Table 2: Comparison with baselines.

dataset was split into a training set (80%), a validation set (10%), and a test set (10%). The detailed distribution of the dataset is shown in Table 1.

For each experiment, we conduct a hyperparameter sweep across learning rate, batch size, and training epochs, selecting the parameters that achieved the highest validation performance. All experiments were performed on a single 4090 GPU. To fine-tune the large language models (LLMs) within the limited memory space of our GPU, we adopted the Low-Rank Adaptation (LoRA) fine-tuning approach (Hu et al., 2021).

In our experiment, we evaluate the results using four metrics: Accuracy, Precision-macro, Recall-macro, and F1-macro.

4.2 Main Results

To fully validate the performance of our method, we select both unimodal and multimodal baselines for comparison.

Specifically, our unimodal baselines encompass both text and image modalities. For the text modality, we select BERT (Devlin et al., 2018), T5 (Raffel et al., 2020), LLaMA-3(LoRa) (Dubey et al., 2024), Xlnet (Yang et al., 2019) and DeBERTa (He et al., 2020) as our baselines. For the image modality, we choose ResNet (He et al., 2016), RepVGG (Ding et al., 2021), and ViT (Dosovitskiy et al., 2020) due to their proven superior performance in image classification tasks. In addition, we also employ some related works including Yuan’s work (Yuan and Lam, 2022), CLMLF (Li et al., 2022b) and BIT (Xiao et al., 2023) as multimodal baselines to provide a comprehensive comparison.

Based on Table 2, we observe that language

models, especially large language models such as LLaMA, perform better in the text modality, highlighting the effectiveness of textual information. Furthermore, the model performance in the vision modality, due to the low density of information, consistently lags behind the text modality, suggesting that images alone are insufficient for sentiment classification in fashion posts. Additionally, multi-modal models do not show a significant improvement over the text modality, possibly due to their limited ability to capture deep correlations between text and images in fashion-related content.

Our proposed model achieves the best results and significantly outperforms all other models ($p < 0.05$). This indicates that our method successfully handle the modality gap though the generation of pseudo data and efficient fusion method.

4.3 Influence of Different MultiModals Integration Methods

Traditional methods typically integrate features from various modalities by directly fusing them and subsequently using a classifier to achieve classification results. Their fusion techniques include concatenation, addition, and cross-attention mechanisms (attention). Additionally, some well-established baseline models such as BLIP (Li et al., 2022a), BLIP2 (Li et al., 2023b), InternLM (Dong et al., 2024), and LLAVA (Liu et al., 2024) have attributed their impressive performance to their specific methods of fusion. In this study, we evaluate these fusion approaches(LoRA finetuned). The results are detailed in Table 3. Notably, for these large multimodal language model, we combined images with pseudo images as the visual input, and

Method	ACC.	P.	R.	F1.
Concatenation	74.72	74.72	75.23	73.47
Addition	73.14	73.07	75.54	71.87
Attention	74.37	74.45	77.23	72.51
BLIP	63.24	68.35	61.19	62.41
BLIP-2	57.08	63.40	62.30	56.90
InternLM	70.69	71.41	73.58	71.20
LLAVA1.5	73.59	73.98	73.58	73.33
Ours	76.64	76.79	77.62	75.07

Table 3: Different multimodal integration methods.

Method	ACC.	P.	R.	F1.
Ours	76.64	76.79	77.62	75.07
-PreTraining	75.49	76.16	76.97	74.42
-Adapter	74.55	75.22	76.00	73.84

Table 4: Ablation studies on proposed multimodal integration method.

text with attributes and pseudo-attributes as the textual input.

The experimental results show that illogical fusing the features from different modalities, regardless of the fusion approach taken, makes it difficult for the model to fully exploit the information embedded in features. The method we adopt, despite its relative complexity, is able to extract information from text and images more logically, efficiently, and accurately, and gives the best results.

We also conduct ablation experiments to demonstrate the effectiveness of the Adapter and the training strategy we use. The results are shown in Table 4.

These results show a decline in model performance when the pre-training strategy is omitted from the training procedure, confirming that pre-training aids our model in distinguishing between images with varying sentiment information before finetuning.

The removal of the Adapter leads to a more substantial decline in all other performance metrics, demonstrating that the Adapter we designed effectively filters out erroneous images, thereby preventing them from adversely affecting the predictions.

4.4 Impact of Different Modalities

In Table 5, we analyze the impact of different modalities, where T , I , and A represent the original review text, images, and attributes, respectively. I_p and A_p denote pseudo images and attributes. In

Method	ACC.	P.	R.	F1.
T	71.24	71.22	71.73	69.43
I	54.66	55.04	54.07	52.14
A	39.74	38.39	37.08	34.27
$T+I$	74.17	74.12	74.30	72.88
$T+A$	74.01	73.01	73.90	71.29
$I+A$	52.57	55.08	50.05	52.72
$I+T+A$	73.85	74.40	75.99	72.22
$I+T+A+I_p$	74.12	75.53	74.96	73.11
$I+T+A+A_p$	74.02	74.02	75.74	72.29
$I+T+A+I_p+A_p$	74.72	74.72	75.23	73.47
Ours	76.64	76.79	77.62	75.07

Table 5: The results of different modalities.

this experiment, features from modalities are fused through concatenating and classified.

From a unimodal perspective, text performs best because it contains clear sentiment information. In contrast, images provide more information but also add noise, making sentiment harder to discern. Attributes, being just supplementary information without inherent sentiment, result in poorer outcomes.

Meanwhile, integrating these modalities yields better results than using them individually, and the utilization of pseudo data leads to further improvements.

Our proposed model significantly outperforms all others, suggesting that integrating various modalities for sentiment classification in fashion posts is beneficial. This also demonstrates the effectiveness of the proposed framework in addressing modality gaps.

4.5 Impact of Pseudo Data Generating Methods

To explore the effectiveness of our pseudo data generating method. We compare different methods in Figure 5. Traditional image rewriting techniques (Figure 5(a-c)) typically rely on basic transformations such as rotation, color modification, and cropping (An et al., 2023). In our study, we design experiments that employ these methods to process images within our dataset, with the original image attributes being preserved as pseudo attributes. The experimental results did not demonstrate a significant improvement over the baseline.

Additionally, numerous studies have employed large-language-model(LLM) to do data rewriting. Inspired by (Dunlap et al., 2023), we explored

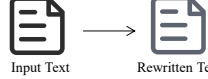
Traditional Methods	Explanation	Results	LLM Methods	Explanation	Results
(a) Rotate 	Rotate the image	ACC.=74.81 P.=74.15 R.=76.41 F1.=73.20	(d) Img2Img 	Get an image description and use it to generate a new image	ACC.=74.46 P.=75.78 R.=73.04 F1.=74.83
(b) Color Change 	Recolor the image on Color, Brightness & Contrast	ACC.=73.91 P.=73.61 R.=75.89 F1.=72.47	(e) Text Rewriting 	Rewrite text without change the meaning	ACC.=73.36 P.=75.05 R.=71.91 F1.=73.85
(c) Cut 	Cut out part of the image	ACC.=74.54 P.=73.36 R.=75.02 F1.=72.02	(f) Ours 	Generate a suitable image description and use it to generate image	ACC.=76.64 P.=76.79 R.=77.62 F1.=75.07

Figure 5: Pseudo data generating methods.

Input	Pseudo Data	Results
 Text: Happy Sunday! On Sundays, we read Here are a list of our Favorite magazines whether digital or print. Attribution: Category: Clothing; Color: grey; Sleeves: short sleeves...	 Pseudo Attribution Description: The person is dressed in a cozy sweater, jeans, and appears content.	Label: Positive Baseline: Neutral Ours: Positive
 Text: Possibly unmotivated picture of Richard Strauss at the age of three. Attribution: Category: Bags\ Backpacks, Material: textile	 Pseudo Attribution Description: He is dressed in oversized, outdated clothes that do not match	Label: Negative Baseline: Neutral Ours: Negative

Figure 6: Examples of case study.

the generation of images using the Image2Image method shown in Figure 5(d). Initially, we generate image descriptions using the vision-language model BLIP-2 (Li et al., 2023b) from the images in our dataset, followed by employing a diffusion model to produce revised images. This approach ranks second to our method, surpassing all other comparative experiments in terms of F1 scores.

Furthermore, drawing on LaCLIP (Fan et al., 2023), we utilized ChatGPT to rewrite text (Figure 5(e)) and subsequently fuse its features with the original information for sentiment prediction but do not get improved results.

4.6 Case Study

In Figure 6, we present a case study comparing our method to a baseline that uses text, image, and attribute features extracted from the CLIP model

for sentiment prediction. We find that in scenarios with complex backgrounds or unclear expressions, the baseline struggles to capture sentiment due to its inability to effectively bridge the modality gap.

In the first example, the baseline struggles with intricate backgrounds, leading to an incorrect prediction. Our approach improves upon this by extracting key information from text and generating a pseudo image with minimized background interference, ensuring a clearer depiction of characters. The pseudo data shows great consistency with its corresponding original data, collectively contributing to more accurate predictions.

In the second example, the word ‘unmotivated’ signals negative sentiment, but is not clearly expressed in the low-resolution image. Our pseudo data captures this and highlights key information in an informative pseudo image, effectively bridging the gap between image and text and yielding correct result compared to the baseline.

5 Conclusion

This paper proposes a novel multimodal framework for sentiment analysis in fashion-related social media, addressing the critical challenge of the modality gap. By generating pseudo data, and utilizing a multimodal fusion approach, the model effectively integrates various modalities to enhance sentiment classification. The self-adaptive adapter ensures the accuracy of pseudo data, contributing to more accurate predictions. Experimental results demonstrate that this framework significantly outperforms both unimodal and multimodal baselines, proving its effectiveness in bridging the modality gap and improving sentiment detection.

6 Limitation

Our dataset includes text, images, and attributes, but the lack of attribute labeling in some sentiment analysis datasets limits the applicability of our method to these datasets. Additionally, the attributes in the dataset we use are AI-labeled, which introduces some errors that could affect the experimental results. Future research will focus on developing more accurate Attribute Value Extraction (AVE) methods to address these challenges.

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